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ACADEMIC DEPARTMENTS

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1. Department of Computer Science and Engineering (CSE)
2. Department of Electronics and Communication Engineering (ECE)
3. Department of Applied Sciences and Humanities (AS&H)

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B.TECH CSE – CURRICULUM OVERVIEW

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Duration: 4 Years (8 Semesters)
Total Credits: ~160 credits

Semester 1 (First Year):

- Mathematics I (Calculus & Linear Algebra)
- Physics / Chemistry (based on group)
- Introduction to Programming (Python / C)
- Engineering Drawing
- Communication Skills (English)
- Environmental Science
- Lab: Programming Lab

Semester 2 (First Year):

- Mathematics II (Discrete Mathematics)
- Physics / Chemistry (alternates from Sem 1)
- Data Structures
- Digital Electronics
- Engineering Workshop / Mechanics
- Lab: Data Structures Lab, Digital Electronics Lab

Semester 3 (Second Year):

- Mathematics III (Probability & Statistics)
- Object-Oriented Programming (Java / C++)
- Computer Organization and Architecture
- Operating Systems
- Database Management Systems (DBMS)
- Labs: OOP Lab, OS Lab, DBMS Lab

Semester 4 (Second Year):

- Design and Analysis of Algorithms
- Computer Networks
- Software Engineering
- Theory of Computation
- Elective I
- Labs: Networking Lab, Project Lab

Semester 5 (Third Year):

- Compiler Design
- Artificial Intelligence
- Machine Learning
- Elective II
- Elective III
- Mini Project

Semester 6 (Third Year):

- Information Security / Cyber Security
- Cloud Computing / Distributed Systems
- Deep Learning
- Elective IV
- Elective V
- Internship / Research Project

Semester 7 (Fourth Year):

- Major Project (Part I)
- Elective VI
- Elective VII
- Seminar

Semester 8 (Fourth Year):

- Major Project (Part II / Final)
- Technical Communication

Popular Electives (CSE):

- Natural Language Processing (NLP)
- Computer Vision
- IoT (Internet of Things)
- Blockchain Technology
- Robotics
- Data Mining & Big Data Analytics
- Game Development

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B.TECH ECE – CURRICULUM OVERVIEW

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Duration: 4 Years (8 Semesters)

Total Credits: ~160 credits

Core Areas:

- Signals and Systems
- Analog and Digital Electronics
- Electromagnetic Theory
- Microprocessors and Microcontrollers
- VLSI Design
- Digital Signal Processing (DSP)
- Communication Systems (Analog + Digital)
- Embedded Systems
- Wireless Communication
- Antenna and Wave Propagation
- Control Systems
- VLSI Testing and Verification

Popular Electives (ECE):

- Robotics and Automation
- 5G Technology
- Nanoelectronics
- Cognitive Radio
- Image and Video Processing
- RF and Microwave Engineering
- Machine Learning for ECE
- Cyber-Physical Systems

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M.TECH PROGRAMS

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M.Tech in CSE – Specialization: Artificial Intelligence

Duration: 2 Years (4 Semesters)

Total Credits: 54 credits

Intake: ~15-20 students per year

Admission:

- Through CCMT (Centralized Counseling for M.Tech/MArch/MPlan/MPharma)
- Requires valid GATE score (CS/DA papers)
- GATE stipend of ₹12,400/month available for eligible students

Curriculum Highlights:

- Advanced Machine Learning
- Deep Learning and Neural Networks
- Computer Vision
- Natural Language Processing

- Reinforcement Learning
- AI in Healthcare / Finance applications
- Research Methodology
- Dissertation (Thesis) – Semester 3 & 4

Research Focus Areas (CSE):

- Artificial Intelligence and Machine Learning
- Natural Language Processing
- Computer Vision and Image Processing
- Cybersecurity and Network Security
- Cloud Computing and IoT
- Data Science and Big Data
- Blockchain and Distributed Systems

M.Tech in ECE – Specialization: Microelectronics & VLSI Design

Duration: 2 Years (4 Semesters)

Total Credits: 54 credits

Intake: ~10-15 students per year

Admission:

- Through CCMT with valid GATE score (EC paper)
- GATE stipend: ₹12,400/month

Curriculum Highlights:

- Advanced VLSI Design
- ASIC Design Flow
- CAD Algorithms for VLSI
- Low-Power VLSI Design
- VLSI Testing and Verification
- Embedded Systems Design
- SoC (System on Chip) Design
- Dissertation (Thesis)

Research Focus Areas (ECE):

- VLSI and Nanoelectronics
- 5G/6G Wireless Communication
- Embedded Systems and IoT
- Robotics and Cyber-Physical Systems
- Antenna Design and EM Waves
- Signal and Image Processing

Ph.D. PROGRAM

Areas: CSE and ECE (same research areas as M.Tech)

Admission Process:

- Conducted TWICE a year: Autumn Semester (July) and Spring Semester (January)
- Step 1: Online Application at iiitp.ac.in
- Step 2: Shortlisting based on GATE / NET / JRF score + academic record
- Step 3: Written Test
- Step 4: Technical Interview with faculty panel
- Step 5: Final merit list and seat allotment

Eligibility:

- M.Tech/M.E./M.S. in relevant field (CSE/ECE/IT/EEE) with 60%+ marks
- Candidates with B.Tech + GATE (exceptional profile) may be considered for direct Ph.D.

Fellowship Schemes:

1. Institute Fellowship: ₹131,000/month (JRF), ₹135,000/month (SRF) – 5 years
2. Visvesvaraya Ph.D. Scheme (MeitY): Additional benefits for IT/electronics research
3. Sponsored: Fee paid by organization; salary as per employment contract
4. Part-time Ph.D.: Available for working professionals

Duration: Minimum 3 years (typically 4-5 years)

FACULTY

IIIT Pune has a growing faculty with Ph.D.s from IITs, IISc, and reputed international universities. Many faculty members:

- Are actively involved in funded research projects (DST, MeitY, SERB, DRDO, ISRO)
- Have publications in IEEE, ACM, Springer, Elsevier journals
- Have industry collaborations with companies like TCS, Infosys, NVIDIA, Qualcomm

Faculty-Student Ratio: Approximately 1:15 (improving as institute grows)

ACADEMIC POLICIES

Attendance:

- Minimum 75% attendance mandatory in each course to appear for exams
- Students with less than 75% may be debarred from semester exams

Grading System:

- 10-point CGPA scale
- Minimum CGPA of 5.0 required to pass / continue
- Minimum CGPA of 6.0 required for placement eligibility in most companies

Examinations:

- Mid-Semester Exam: ~30% weightage
- End-Semester Exam: ~50% weightage
- Assignments/Labs/Quizzes: ~20% weightage

Backlogs:

- Students can re-appear in failed courses in subsequent semesters
- Maximum attempts allowed: Usually 3 attempts per course

RESEARCH AND INNOVATION

Research Centers:

- Center for AI and Data Science
- Center for Embedded Systems and VLSI
- Innovation Lab (Startup Incubation Support)

Funded Projects:

- Projects from DST (Department of Science and Technology)
- MeitY (Ministry of Electronics and IT) funded projects
- DRDO collaborations
- Industry-sponsored research

Publications:

- Students and faculty publish in IEEE Transactions, ACM conferences, Springer journals

National/International Competitions:

- Students participate in Smart India Hackathon (SIH)
- ACM-ICPC (competitive programming)
- IEEE conferences and paper presentations
- DRDO/ISRO technical competitions